
CS246: Database Management Systems Lab

Week # 02 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 13-Jan-2025 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 1

Submission time: 16:45 Hrs, 13-Jan-2025

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Question 1: (100 points)

Sort the input records by specified set of keys. Implement the following problem in C programming language. Use of any other programming language is not allowed and will lead to awarding 0 marks.

Task 01 (2 marks) Declare a structure `time` containing two elements namely

1. hrs `int`
2. mins `int`

Task 02 (5 marks) Declare a structure `record` containing five elements

1. Origin `char(3)`
2. Destination `char(3)`
3. Flight number `char(6)`
4. Departure time `struct time`
5. Arrival time `struct time`

Task 03 (3 marks) Read input file `week02-database.txt` of the following format into the `record` structure.

```
BOM HYD 6E5056 21:30 22:55
BOM GAU 6E228 08:40 11:30
...
```

Task 04 (80 marks) Sort the data on the following fields using Quick sort

1. (20 marks) Quick sort implementation carry 20 marks
2. (20 marks) Origin airport in ascending order
3. (20 marks) Followed by destination airport in descending order
4. (20 marks) Followed by departure time in ascending order

Task 05 (10 marks) Output at the end of Task 04 should be written in a new file with the each line format identical to the input file format